

Lecture 13: Fast Reinforcement Learning

Typo: Lecture 11

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CS234 Reinforcement Learning

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- With some slides from or derived from David Silver, Examples new

Refresh Your Understanding: Deterministic Bandits

- Which of the following statements is true in a bandit problem with deterministic rewards?
 - ① Upper Confidence Bound (UCB) will have sublinear regret with probability one
 - ② Upper Confidence Bound (UCB) will have linear regret with some strictly positive probability
 - ③ a division by zero will occur in the Upper Confidence Bound (UCB) algorithm and so it cannot handle deterministic environments
 - ④ Upper Confidence Bound (UCB) will always exhibit more regret than in a stochastic environment because it will over-explore.
 - ⑤ Not Sure

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 - 3 a division by zero will occur in the Upper Confidence Bound (UCB) algorithm and so it cannot handle deterministic environments
 - 4 Upper Confidence Bound (UCB) will always exhibit more regret than in a stochastic environment because it will over-explore.
 - 5 Not Sure

Class Survey Results

- Thank you to everyone that filled it in!
- The Good: Tutorials (+40), Check your understandings & lectures and math
- Things to Improve: Lectures and math (want more high level structure and conceptual understanding and examples)
- What I Will Change: Will try to emphasize conceptual ideas and provide concrete examples

Where We are

- Last time: Bandits and regret and UCB (fast learning)
- This time: Bandits, regret and Bayesian bandits (fast learning)

Today

- Bayesian bandits
- Thompson sampling
- Bayesian Regret

Multiarmed Bandits Notation Recap

- Multi-armed bandit is a tuple of $(\mathcal{A}, \mathcal{R})$
- $\mathcal{A}$: known set of m actions (arms)
- $\mathcal{R}^a(r) = \mathbb{P}[r \mid a]$ is an unknown probability distribution over rewards
- At each step t the agent selects an action $a_t \in \mathcal{A}$
- The environment generates a reward $r_t \sim \mathcal{R}^{a_t}$
- Goal: Maximize cumulative reward $\sum_{\tau=1}^t r_\tau$
- **Regret** is the opportunity loss for one step

$$l_t = \mathbb{E}[V^* - Q(a_t)]$$

- **Total Regret** is the total opportunity loss

$$L_t = \mathbb{E}\left[\sum_{\tau=1}^t V^* - Q(a_\tau)\right]$$

- Maximize cumulative reward $\iff$ minimize total regret

- **Bayesian bandits**
- Thompson Sampling
- Bayesian Regret

- So far we have made no assumptions about the reward distribution $\mathcal{R}$
 - Except bounds on rewards
- **Bayesian bandits** exploit prior knowledge of rewards, $p[\mathcal{R}]$

Short Refresher on / Introduction to Bayesian Inference

- In Bayesian view, we start with a prior over the unknown parameters
 - Here the unknown distribution over the rewards for each arm
- Given observations / data about that parameter, update our uncertainty over the unknown parameters using Bayes Rule

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- For example, let the reward of arm i be a probability distribution that depends on parameter ϕ_i
- Initial prior over ϕ_i is $p(\phi_i)$
- Pull arm i and observe reward r_{i1}
- Use Bayes rule to update estimate over ϕ_i :

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- Use Bayes rule to update estimate over ϕ_i :

$$p(\phi_i | r_{i1}) = \frac{p(r_{i1} | \phi_i) p(\phi_i)}{p(r_{i1})} = \frac{p(r_{i1} | \phi_i) p(\phi_i)}{\int_{\phi_i} p(r_{i1} | \phi_i) p(\phi_i) d\phi_i}$$

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$$p(\phi_i | r_{i1}) = \frac{p(r_{i1} | \phi_i) p(\phi_i)}{\int_{\phi_i} p(r_{i1} | \phi_i) p(\phi_i) d\phi_i}$$

- In general computing this update may be tricky to do exactly with no additional structure on the form of the prior and data likelihood

Short Refresher on / Introduction to Bayesian Inference: Conjugate

- In Bayesian view, we start with a prior over the unknown parameters
- Give observations / data about that parameter, update our uncertainty over the unknown parameters using Bayes Rule

$$p(\phi_i | r_{i1}) = \frac{p(r_{i1} | \phi_i) p(\phi_i)}{\int_{\phi_i} p(r_{i1} | \phi_i) p(\phi_i) d\phi_i}$$

- In general computing this update may be tricky
- But sometimes can be done analytically
- If the parametric representation of the prior and posterior is the same, the prior and model are called **conjugate**
- For example, exponential families have conjugate priors

Short Refresher on / Introduction to Bayesian Inference: Bernoulli

- Consider a bandit problem where the reward of an arm is a binary outcome 0, 1, sampled from a Bernoulli with parameter θ
 - E.g. Advertisement click through rate, patient treatment success/fails, ...
- The Beta distribution $Beta(\alpha, \beta)$ is conjugate for the Bernoulli distribution

$$p(\theta|\alpha, \beta) = \theta^{\alpha-1}(1-\theta)^{\beta-1} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$$

where $\Gamma(x)$ is the Gamma family

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where $\Gamma(x)$ is the Gamma family

- Assume the prior over θ is $Beta(\alpha, \beta)$ as above
- Then after observed a reward $r \in \{0, 1\}$ then updated posterior over θ is $Beta(r + \alpha, 1 - r + \beta)$

Bayesian Inference for Decision Making

- Maintain distribution over reward parameters
- Use this to inform action selection

Bayesian bandits Overview

- So far we have made no assumptions about the reward distribution $\mathcal{R}$
 - Except bounds on rewards
- **Bayesian bandits** exploit prior knowledge of rewards, $p[\mathcal{R}]$
- They compute posterior distribution of rewards $p[\mathcal{R} \mid h_t]$, where $h_t = (a_1, r_1, \dots, a_{t-1}, r_{t-1})$
- Use posterior to guide exploration
 - Upper confidence bounds (Bayesian UCB)
 - Probability matching (Thompson Sampling)
- Better performance if prior knowledge is accurate

Today

- Bandits and Probably Approximately Correct
- Bayesian bandits
- Thompson Sampling
- Bayesian Regret

Thompson Sampling

- 1: Initialize prior over each arm a , $p(\mathcal{R}_a)$
- 2: **for** iteration= $1, 2, \dots$ **do**
- 3: For each arm a **sample** a reward distribution $\mathcal{R}_a$ from posterior
- 4: Compute action-value function $Q(a) = \mathbb{E}[\mathcal{R}_a]$
- 5: $a_t = \arg \max_{a \in \mathcal{A}} Q(a)$
- 6: Observe reward r
- 7: Update posterior $p(\mathcal{R}_a)$ using Bayes Rule
- 8: **end for**

Toy Example: Ways to Treat Broken Toes, Thompson Sampling

- True (unknown) Bernoulli parameters for each arm/action
- Surgery: $\theta_1 = .95$ / Taping: $\theta_2 = .9$ / Nothing: $\theta_3 = .1$
- Thompson sampling:
- Place a prior over each arm's parameter. Here choose Beta(1,1) (Uniform)
 - 1 Sample a Bernoulli parameter given current prior over each arm Beta(1,1), Beta(1,1), Beta(1,1):



Toy Example: Ways to Treat Broken Toes, Thompson Sampling¹

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- Thompson sampling:
- Place a prior over each arm's parameter. Here choose Beta(1,1)
 - 1 Sample a Bernoulli parameter given current prior over each arm
Beta(1,1), Beta(1,1), Beta(1,1): 0.3 0.5 0.6
 - 2 Select $a = \arg \max_{a \in A} Q(a) = \arg \max_{a \in A} \theta(a) =$

¹Note: This is a made up example. This is not the actual expected efficacies of the various treatment options for a broken toe

Toy Example: Ways to Treat Broken Toes, Thompson Sampling

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- Surgery: $\theta_1 = .95$ / Taping: $\theta_2 = .9$ / Nothing: $\theta_3 = .1$
- Thompson sampling:
- Place a prior over each arm's parameter. Here choose $\theta_i \sim \text{Beta}(1,1)$
 - 1 Per arm, sample a Bernoulli θ given prior: 0.3 0.5 0.6
 - 2 Select $a_t = \arg \max_{a \in A} Q(a) = \arg \max_{a \in A} \theta(a) = 3$
 - 3 Observe the patient outcome's outcome: 0
 - 4 Update the posterior over the $Q(a_t) = Q(a^3)$ value for the arm pulled

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 - 1 Sample a Bernoulli parameter given current prior over each arm
 $\text{Beta}(1,1), \text{Beta}(1,1), \text{Beta}(1,1): 0.3 \ 0.5 \ 0.6$
 - 2 Select $a_t = \arg \max_{a \in A} Q(a) = \arg \max_{a \in A} \theta(a) = 3$
 - 3 Observe the patient outcome's outcome: 0
 - 4 Update the posterior over the $Q(a_t) = Q(a^1)$ value for the arm pulled
 - $\text{Beta}(c_1, c_2)$ is the conjugate distribution for Bernoulli
 - If observe 1, $c_1 + 1$ else if observe 0 $c_2 + 1$
 - 5 New posterior over Q value for arm pulled is:
 - 6 New posterior $p(Q(a^3)) = p(\theta(a_3) = \text{Beta}(1,2)$

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 - 1 Sample a Bernoulli parameter given current prior over each arm
Beta(1,1), Beta(1,1), Beta(1,1): 0.3 0.5 0.6
 - 2 Select $a_t = \arg \max_{a \in A} Q(a) = \arg \max_{a \in A} \theta(a) = 1$
 - 3 Observe the patient outcome's outcome: 0
 - 4 New posterior $p(Q(a^1)) = p(\theta(a_1) = \text{Beta}(1, 2)$

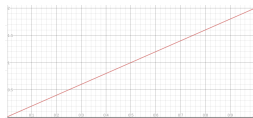


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- True (unknown) Bernoulli parameters for each arm/action
- Surgery: $\theta_1 = .95$ / Taping: $\theta_2 = .9$ / Nothing: $\theta_3 = .1$
- Thompson sampling:
- Place a prior over each arm's parameter. Here choose $\theta_i \sim \text{Beta}(1,1)$
 - ① Sample a Bernoulli parameter given current prior over each arm
Beta(1,1), Beta(1,1), Beta(1,2): 0.7, 0.5, 0.3

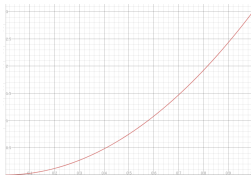
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- Place a prior over each arm's parameter. Here choose $\theta_i \sim \text{Beta}(1,1)$
 - 1 Sample a Bernoulli parameter given current prior over each arm
Beta(1,1), Beta(1,1), Beta(1,2): 0.7, 0.5, 0.3
 - 2 Select $a_t = \arg \max_{a \in A} Q(a) = \arg \max_{a \in A} \theta(a) = 1$
 - 3 Observe the patient outcome's outcome: 1
 - 4 New posterior $p(Q(a^1)) = p(\theta(a_1) = \text{Beta}(2, 1)$



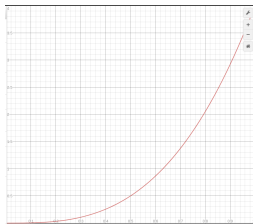
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- Thompson sampling:
- Place a prior over each arm's parameter. Here choose $\theta_i \sim \text{Beta}(1,1)$
 - 1 Sample a Bernoulli parameter given current prior over each arm
Beta(2,1), Beta(1,1), Beta(1,2): 0.71, 0.65, 0.1
 - 2 Select $a_t = \arg \max_{a \in A} Q(a) = \arg \max_{a \in A} \theta(a) = 1$
 - 3 Observe the patient outcome's outcome: 1
 - 4 New posterior $p(Q(a^1)) = p(\theta(a_1) = \text{Beta}(3, 1)$



Toy Example: Ways to Treat Broken Toes, Thompson Sampling

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- Surgery: $\theta_1 = .95$ / Taping: $\theta_2 = .9$ / Nothing: $\theta_3 = .1$
- Thompson sampling:
- Place a prior over each arm's parameter. Here choose $\theta_i \sim \text{Beta}(1,1)$
 - 1 Sample a Bernoulli parameter given current prior over each arm
Beta(2,1), Beta(1,1), Beta(1,2): 0.75, 0.45, 0.4
 - 2 Select $a_t = \arg \max_{a \in A} Q(a) = \arg \max_{a \in A} \theta(a) = 1$
 - 3 Observe the patient outcome's outcome: 1
 - 4 New posterior $p(Q(a^1)) = p(\theta(a_1)) = \text{Beta}(4, 1)$



Toy Example: Ways to Treat Broken Toes, Thompson Sampling vs Optimism

- Surgery: $\theta_1 = .95$ / Taping: $\theta_2 = .9$ / Nothing: $\theta_3 = .1$
- How does the sequence of arm pulls compare in this example so far?

Optimism	TS
a^1	a^3
a^2	a^1
a^3	a^1
a^1	a^1
a^2	a^1

On to General Setting for Thompson Sampling

- Now we will see how Thompson sampling works in general, and what it is doing

Probability Matching

- Assume have a parametric distribution over rewards for each arm
- **Probability matching** selects action a according to probability that a is the optimal action

$$\pi(a \mid h_t) = \mathbb{P}[Q(a) > Q(a'), \forall a' \neq a \mid h_t]$$

- Can be difficult to compute probability that an action is optimal analytically from posterior
- Somewhat incredibly, Thompson sampling implements probability matching

Thompson sampling implements probability matching

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- Thompson sampling:

$$\begin{aligned}\pi(a \mid h_t) &= \mathbb{P}[Q(a) > Q(a'), \forall a' \neq a \mid h_t] \\ &= \mathbb{E}_{\mathcal{R} \mid h_t} \left[\mathbb{1}(a = \arg \max_{a \in \mathcal{A}} Q(a)) \right]\end{aligned}$$

Probability Matching

- Assume have a parametric distribution over rewards for each arm
- **Probability matching** selects action a according to probability that a is the optimal action

$$\pi(a \mid h_t) = \mathbb{P}[Q(a) > Q(a'), \forall a' \neq a \mid h_t]$$

- Can be difficult to compute probability that an action is optimal analytically from posterior
- Somewhat incredibly, Thompson sampling implements probability matching
- Note: probability matching is often optimistic in the face of uncertainty
 - Uncertain actions have higher probability of being max

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Framework: Regret and Bayesian Regret

- How do we evaluate performance in the Bayesian setting?
- Frequentist regret assumes a true (unknown) set of parameters

$$\text{Regret}(\mathcal{A}, T; \theta) = \mathbb{E}_{\tau} \left[\sum_{t=1}^T Q(a^*) - Q(a_t) \mid \theta \right]$$

where $\mathbb{E}_{\tau}$ denotes an expectation with respect to the history of actions taken and rewards observed given an algorithm $\mathcal{A}$.

- Bayesian regret assumes there is a prior over parameters

$$\text{BayesRegret}(\mathcal{A}, T; \theta) = \mathbb{E}_{\theta \sim p_{\theta}, \tau} \left[\sum_{t=1}^T Q(a^*) - Q(a_t) \mid \theta \right]$$

Bounding Regret Using Optimism

- How do we evaluate performance in the Bayesian setting?
- Frequentist regret assumes a true (unknown) set of parameters

$$\text{Regret}(\mathcal{A}, T; \theta) = \mathbb{E}_{\tau} \left[\sum_{t=1}^T Q(a^*) - Q(a_t) | \theta \right] \leq \mathbb{E}_{\tau} \left[\sum_{t=1}^T U_t(a_t) - Q(a_t) | \theta \right]$$

where $\mathbb{E}_{\tau}$ denotes an expectation with respect to the history of actions taken and rewards observed given an algorithm $\mathcal{A}$ (under event that U_t is an upper bound).

Thompson sampling

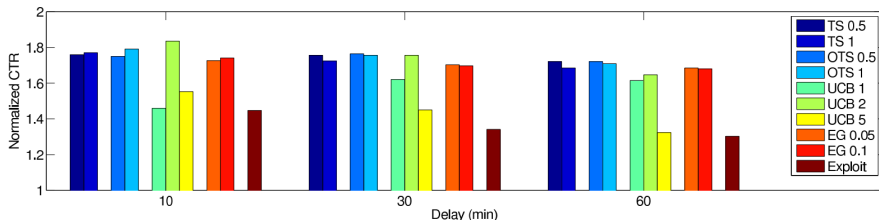
- Frequentist bounds for standard* Thompson sampling do not* (last checked) match best bounds for frequentist algorithms
- Empirically Thompson sampling can be effective, especially in contextual multi-armed bandits

Contextual Bandit Thompson Sampling

- Contextual bandit: input context which impacts reward of each arm, context sampled iid each step
- Arms = articles
- Reward = click (+1) on article ($Q(a)$ =click through rate)

Thompson Sampling for News Article Recommendation (Chapelle and Li, 2010)

- Contextual bandit: input context which impacts reward of each arm, context sampled iid each step
- Arms = articles
- Reward = click (+1) on article ($Q(a)$ =click through rate)



Check Your Understanding: Thompson Sampling and Optimism

- Consider an online news website with thousands of people logging on each second. Frequently a new person will come online before we see whether the last person has clicked (or not). Select all that are true:
 - 1 Thompson sampling would be better than optimism here, because optimism algorithms are deterministic and would select the same action until we get feedback (click or not)
 - 2 Optimism algorithms would be better than TS here, because they have stronger regret bounds for this setting
 - 3 Thompson sampling could cause much worse performance than optimism if the initial prior is very misleading.
 - 4 Not sure

Check Your Understanding: Thompson Sampling and Optimism **Solutions**

- Consider an online news website with thousands of people logging on each second. Frequently a new person will come online before we see whether the last person has clicked (or not). Select all that are true:
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 - ③ Thompson sampling could cause much worse performance than optimism if the initial prior is very misleading.
 - ④ Not sure

Optimal Policy for Bayesian bandits?

- Thompson Sampling often works well, but is it optimal?
- Given prior, and known horizon, could compute decision policy that would maximize expected rewards given the available horizon
- Computational challenge: naively this would create a decision policy that is a function of the history to the next arm to pull

Gittins Index for Bayesian bandits

- Thompson Sampling often works well, but is it optimal?
- Given prior, and known horizon, could compute decision policy that would maximize expected rewards given the available horizon
- Computational challenge: naively this would create a decision policy that is a function of the history to the next arm to pull
- **Index policy**: a decision policy that computes a "real-valued index for each arm and plays the arm with the largest index," using statistics only from that arm and the horizon (definition from Lattimore and Svespari 2019 Bandit Algorithms)
- **Gittins index**: optimal policy for maximizing expected discounted reward in a Bayesian multi-armed bandit

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What You Should Understand

- Understand how multi-armed bandits relate to MDPs
- Be able to define regret and PAC
- Be able to prove why UCB bandit algorithm has sublinear regret
- Understand (be able to give an example) why e-greedy and greedy and pessimism can result in linear regret
- Be able to implement UCB bandit algorithm
- Be able to implement Thompson Sampling for Bernoulli rewards

Where We are

- Last time: Bandits and regret and UCB (fast learning)
- This time: Bayesian bandits (fast learning)

Bayesian Regret Bounds for Thompson Sampling

- $\text{Regret}(\text{UCB}, T)$

$$\text{BayesRegret}(TS, T) = E_{\theta \sim p_\theta} \left[\sum_{t=1}^T f^*(a^*) - f^*(a_t) \right]$$

- Posterior sampling has the same (ignoring constants) regret bounds as UCB

Toy Example: Probably Approximately Correct and Regret

- Surgery: $\phi_1 = .95$ / Taping: $\phi_2 = .9$ / Nothing: $\phi_3 = .1$
- Let $\epsilon = 0.05$
- O = Optimism, TS = Thompson Sampling: W/in
 $\epsilon = \mathbb{I}(Q(a_t) \geq Q(a^*) - \epsilon)$

O	TS	Optimal	O Regret	O W/in ϵ	TS Regret	TS W/in ϵ
a^1	a^3	a^1	0	Y	0.85	N
a^2	a^1	a^1	0.05	Y	0	Y
a^3	a^1	a^1	0.85	N	0	Y
a^1	a^1	a^1	0	Y	0	Y
a^2	a^1	a^1	0.05	Y	0	Y